

Arithmetic Hacks

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1 2 digit number divided by 2

$$([a][b])/2$$

$$a \text{ even} \Rightarrow [\frac{a}{2}][\frac{b}{2}]$$

$$b \text{ even} \Rightarrow [\frac{a-1}{2}][\frac{10+b}{2}]$$

2 examples

$$26/2 = [2/2][6/2] = 13$$

$$34/2 = [\frac{3-1}{2}][\frac{10+4}{2}] = [\frac{2}{2}][\frac{14}{2}] = 17$$

3 $[a][9] \times 3$

$$[a][9] \times 3 = [(a+1) \times 3][0] - b$$

4 examples

$$19 \times 3 = [(1+1) \times 3][0] - 3 = 60 - 3 = 57$$

$$29 \times 3 = [(2+1) \times 3][0] - 3 = 90 - 3 = 87$$

5 modular division of many digit numbers

modular division can be done blockwise,

here is an example

$$178247\%7 = [17\%7][8\%7][247\%7]$$

(note block sizes are arbitrary)

$$[17\%7][8\%7][247\%7] = [3][1][002]$$

$$[31\%7][002]$$

$$[3][002]$$

$$[30\%7][02]$$

$$[2][02]$$

$$202\%7 = 6$$

$$\text{therefore } 178247\%7 = 6$$

reference: Euler